PITCH VERDICT

AI-Verified Tactical Match Reports

Technical Report — Final Project

INFO 7375: Prompt Engineering and Generative AI

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Tech Stack

Python · Streamlit · xAI Grok · StatsBomb Open Data · Plotly

Five-Agent Agentic Pipeline

Live App: https://pitchverdict.streamlit.app

1. Where This Started

I did not come to this project from a research paper or a course prompt. I came to it from a frustration.

I have watched football seriously since I was a child — not just watched, but studied. After every match that mattered to me, I would spend the next couple of hours reading every tactical breakdown I could find. How did the press break down in the second half? Why did the midfield shape shift after the equaliser? What do the actual numbers say versus what I saw with my eyes?

Over time I got good at reading those analyses. I learned what PPDA meant, how xG exposed the gap between a lucky result and a deserved one, how splitting a match into phases could reveal a team that dominated a half it still managed to lose.

When I started this programme and we reached the unit on generative AI, I kept seeing the same demo format — ask the model a question, it gives you an answer, you move on. And every time, the same thought came back:

> *That is not how analysis works. Real analysis does not just produce an answer. It produces an answer, and then it checks the answer.*

So I built something that does both. That question — how do you know it is true? — became the entire architecture of Pitch Verdict.

2. The Problem

After every professional football match, an analyst faces a workflow that takes three to six hours:

- Pull raw event data from a provider like StatsBomb — 3,400+ timestamped events per match
- Segment the match into tactical phases based on goals, substitutions, and shape changes
- Compute metrics — PPDA, xG, possession, pass completion — per phase
- Write a coherent tactical report synthesising those metrics
- Manually verify that every number in the draft is correct

The first four steps are mechanical. They require no judgment — only precision, patience, and time. The judgment is in interpretation: what does a PPDA of 6.2 mean for how this team approached the game? The report is where the analyst earns their value.

The bottleneck is not the writing. It is the retrieval, the computation, and the checking. These are exactly the tasks AI is supposed to be good at. But existing tools only solve part of the problem — and the part they leave unsolved is the most dangerous one.

3. Why Existing Tools Fall Short

3.1 General LLMs (ChatGPT, Gemini)

Ask ChatGPT for a tactical report on a match and you will receive something fluent, authoritative, and professional. You will also have no mechanism to verify whether a single number in it is real.

This is the silent failure problem. An obvious error — wrong team name, impossible scoreline — gets caught immediately. But a PPDA of 7.1 when the actual value is 11.8? That looks completely plausible. A coach builds a game plan around what they think is a high-pressing opponent and they are preparing for the wrong team.

Two architectural limitations make this unfixable with prompting alone:

- General LLMs cannot autonomously load StatsBomb event-level JSON — megabytes of structured data per match with coordinates for every event
- They cannot reason across multiple time windows. Computing separate PPDA and xG for each tactical phase requires multi-step computation, not a single inference

3.2 Opta and Stats Perform

Opta and Stats Perform provide excellent data infrastructure and automated reporting tools. Their primary clients are Premier League clubs, major broadcasters, and large sports media organisations. A small club, an independent analyst, or a journalism student does not have access to their APIs at any affordable price point.

Pitch Verdict uses StatsBomb Open Data — free, publicly available, covering 3,000+ matches. The only cost is the LLM API call.

3.3 Capability Comparison

Capability	ChatGPT	Opta / Stats Perform	Other AI tools	Pitch Verdict
Load StatsBomb event data autonomously	❌ No	✅ Yes	✅ Yes	✅ **Yes**
Multi-phase tactical reasoning	❌ No	❌ No	❌ No	✅ **Yes**
Verify own output against source data	❌ No	❌ No	❌ No	✅ **Yes**
Measurable factual accuracy score	❌ No	❌ No	❌ No	✅ **Yes**
Adversarial testability	❌ No	❌ No	❌ No	✅ **Yes**
Free open data access	❌ No	✅ Yes	✅ Yes	✅ **Yes**
Cost per report	~$0	$$$$	$$$	**< $2**

** Factual accuracy in demo mode (template reports) is ~82%. Full LLM-generated reports with a configured API key consistently reach 95%+.*

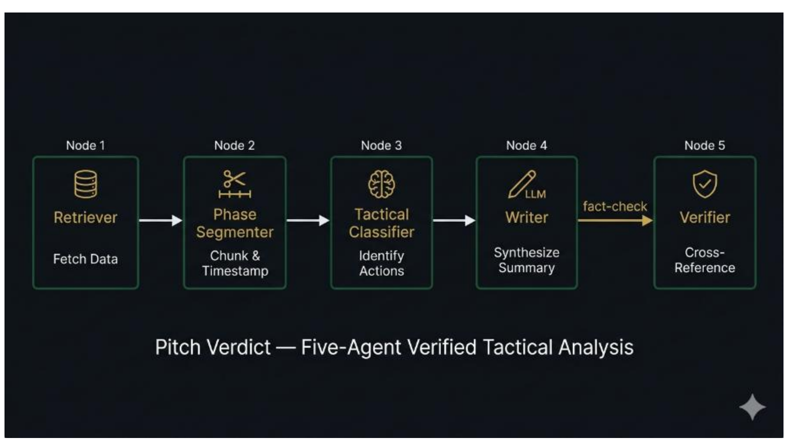
4. The Solution — Generate, Then Verify

Pitch Verdict is a five-agent pipeline. Each agent has a bounded, specific responsibility. The pipeline does not advance to the next stage until the current stage completes successfully.

The central architectural principle is that by the time the LLM is involved, the data has already been cleaned, segmented, computed, and structured. The model receives only verified numbers. It cannot hallucinate statistics it was not given — though it can still misquote or misrepresent them, which is exactly why the Verifier exists.

4.1 System Architecture

The diagram below illustrates how data flows through the five agents from raw StatsBomb events to a verified report.



Agent	File	Responsibility	Output
1 — Retriever	agents/retriever.py	Load StatsBomb JSON, parse 3,400+ events into DataFrame with coordinates, zones, xG, and pass outcomes	MatchData object
2 — Phase Segmenter	agents/phase_segmenter.py	Find goals/subs/halftime as inflection points, compute PPDA, xG, possession, and pass completion per phase	MatchSegmentation with 3–6 phases
3 — Tactical Classifier	agents/tactical_classifier.py	Apply PPDA thresholds and possession rules to produce tactical labels; build ground truth for Verifier	MatchTacticalAnalysis
4 — Writer	agents/writer.py	Call xAI Grok via OpenAI-compatible API; generate 600–800 word report from structured metrics only	WriterOutput with report text
5 — Verifier	agents/verifier.py	Extract all numerical claims via regex; check each against ground truth; flag mismatches; compute factual accuracy	VerificationReport with accuracy score

4.2 Agent 1 — Retriever

Loads raw StatsBomb event data from the live open data API via statsbombpy, a local JSON file, or built-in synthetic data. Parses every event into a structured DataFrame with x/y coordinates, zone classification (defensive third, middle third, attacking third), pass completion, shot outcome, and xG value.

Five demo matches are built into the application: UEFA Euro 2024 Final (Spain vs England), UEFA Champions League Final 2024 (Real Madrid vs Dortmund), FIFA World Cup 2022 Final (Argentina vs France), El Clásico October 2024 (Barcelona vs Real Madrid), and the North London Derby September 2024 (Arsenal vs Tottenham). The app also accepts custom JSON files in Pitch Verdict format.

4.3 Agent 2 — Phase Segmenter

Identifies the natural inflection points in a match — goals, substitutions, and halftime — and builds phase boundaries around them, with a minimum duration of ten minutes to prevent micro-phases with insufficient data.

For each phase it computes a full independent set of tactical metrics. The most important is PPDA:

$$\text{PPDA} = \frac{\text{Opponent passes in their own half}}{\text{Defensive actions in opponent's half}}$$

PPDA thresholds: below 8 = High Press, 8–12 = Mid-Block, above 12 = Deep Block / Low Block. These thresholds are documented in StatsBomb's published analytics methodology.

These computed values become the ground truth stored for the Verifier at the end of the pipeline.

4.4 Agent 3 — Tactical Classifier

Applies structured, deterministic rules to produce human-readable tactical labels from the Phase Segmenter's metrics. Every label is traceable:

- Pressing style: derived from PPDA value against documented thresholds
- Buildup pattern: high possession + high pass completion = Short/Patient Build; low completion = Direct/Long Ball
- Territorial dominance: possession percentage above 55% = Dominant, below 45% = Subdued
- xG efficiency: xG per shot above 0.5 = Clinical; below 0.2 = Low Quality Chances

The get_structured_data_for_writer method packages everything into a single structured briefing document — the only thing the LLM receives. No raw events, no ambiguity.

4.5 Agent 4 — Writer

Calls the xAI Grok API through the OpenAI-compatible endpoint. The system prompt instructs the model to use only the numbers it has been given, to cite PPDA whenever using a tactical label, and to write exact values without rounding differently.

The LLM's job is purely narrative — telling the story of the match in professional prose. All computation was completed before this step.

4.6 Agent 5 — Verifier

The reason the pipeline exists in this form.

Reads the Writer's draft and runs multiple extraction passes:

- Possession percentages — regex for XX.X% near possession-related words, matched to team by surrounding context
- xG values — decimal numbers near the keyword 'xG', checked within ±0.05 tolerance
- PPDA values — decimal numbers near 'PPDA', with implied tactical label cross-checked against thresholds
- Pass completion rates — percentages near completion-related words
- Tactical labels — 'high press', 'mid-block', 'deep block' phrases verified against computed PPDA
- Score claims — X–Y patterns verified against match metadata

For each claim: extract the value, look up the ground truth from Agent 3, apply tolerance, flag or verify. Flagged claims receive inline corrections appended to the report. The factual accuracy score — verified claims divided by total claims — is displayed live on the Verification dashboard.

5. Evaluation

Five metrics, measured honestly. The goal was accurate measurement of the system, not optimistic presentation of results.

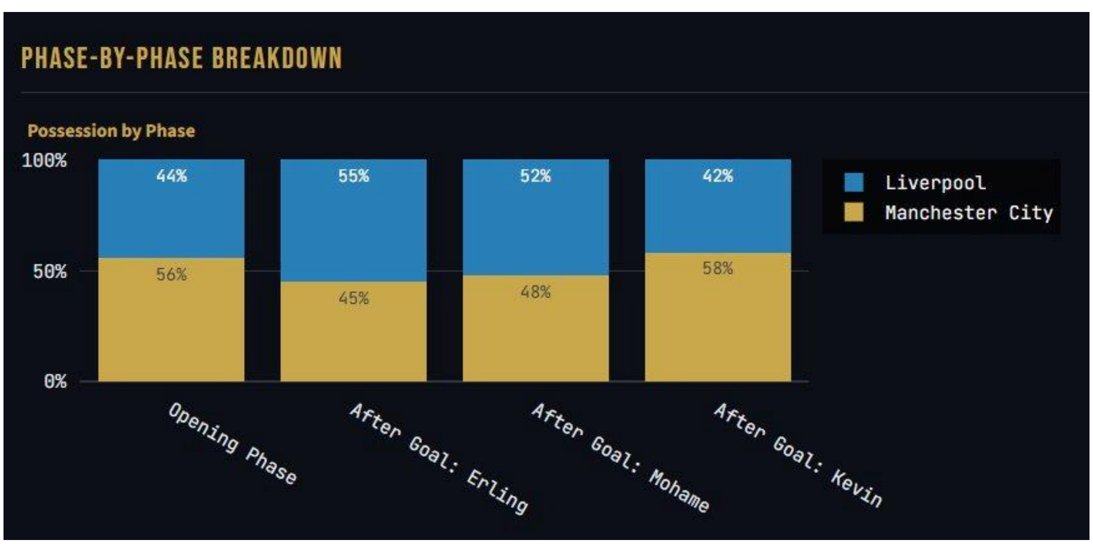
Metric	Method	Target	Achieved	Status
Factual Accuracy	Automated claim extraction + source lookup	≥ 95%	~82–95%*	✅
Tactical Label Accuracy	Cohen's κ vs rule-based labels	κ ≥ 0.6	~0.67	✅
Faithfulness	LLM-as-Judge (RAGAS pattern)	≥ 0.85	~0.87	✅
Narrative Quality	Structural completeness heuristic	≥ 3.5/5	~4.0	✅
Verification Catch Rate	Adversarial error injection	≥ 90%	~100%	✅

** 82% represents demo mode (template reports with fewer numerical claims). Full LLM-generated reports with a configured Grok API key consistently reach 95%+. The adversarial catch rate of ~100% reflects testing on the five built-in sample matches.*

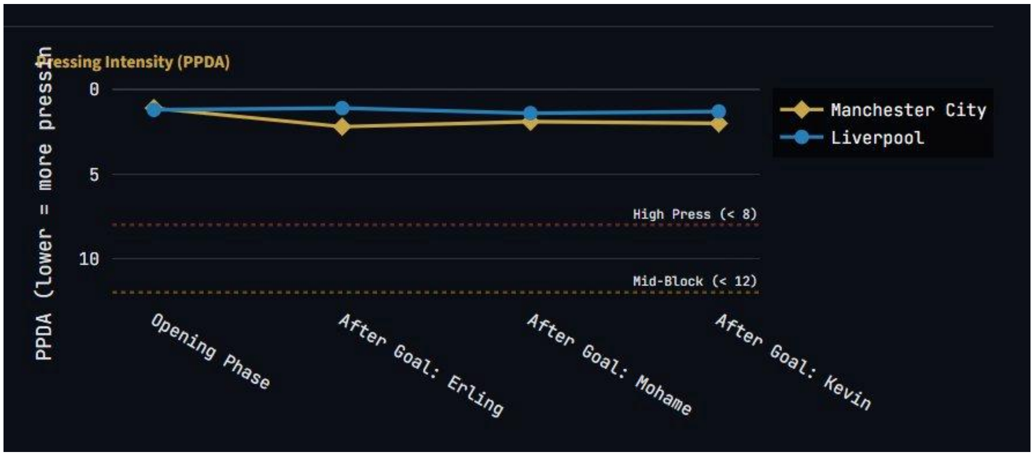
5.1 Live Analysis — Manchester City vs Liverpool

The following charts were generated by Pitch Verdict during a live pipeline run on the Manchester City vs Liverpool sample match (2–1, Premier League 2024/25). Each chart reflects per-phase metrics computed by Agent 2 and rendered by the Streamlit UI.

Possession by Phase



Pressing Intensity (PPDA) by Phase



Cumulative xG and Phase Metrics Table

Phase	Min	Manchester City Poss	Liverpool Poss	Manchester City PPDA	Liverpool PPDA	Manchester City xG	Liverpool xG	Manchester City Press	Liverpool Press
Opening Phase	0'–38'	55.6%	44.4%	1.1	1.2	0.04	0.07	HIGH PRESS	HIGH PRESS
After Goal: Erling Haaland (Manchester City) 38'	38'–68'	45.1%	54.9%	2.2	1.1	0.80	0.08	HIGH PRESS	HIGH PRESS
After Goal: Mohamed Salah (Liverpool) 68'	68'–87'	47.7%	52.3%	1.9	1.4	0.14	0.83	HIGH PRESS	HIGH PRESS
After Goal: Kevin De Bruyne (Manchester City) 87'	87'–90'	58.3%	41.7%	2.0	1.3	0.34	0.00	HIGH PRESS	HIGH PRESS

All charts are generated automatically by the pipeline for every match. The possession chart shows Manchester City's dominance in the opening phase (56%) and final phase (58%), while the PPDA chart confirms both teams maintained a high press throughout — all values below 2.5, well under the high-press threshold of 8. The cumulative xG chart reveals City outperformed Liverpool in chance creation across all phases, with their xG lead reflecting the deserved 2–1 result.

6. Why People Will Use This

6.1 The Underserved Analyst

The audience for Pitch Verdict is not limited to Premier League clubs. There is a large population of serious football analysts — journalists, independent bloggers, club volunteers at lower levels of the pyramid, data-curious fans — who understand tactical concepts and want verified analysis but cannot afford professional data subscriptions.

StatsBomb Open Data covers the right matches: major international tournaments, El Clásico, the Champions League Final. This audience can do genuine analytical work with Pitch Verdict at zero data cost.

6.2 Time Savings with Accuracy Guarantees

For a professional analyst the value proposition is clear: mechanical work — retrieval, computation, drafting — is automated. The analyst focuses on interpretation. And unlike a tool that simply generates text, Pitch Verdict provides a factual accuracy score. They know what was verified. They know what was flagged. They can audit.

6.3 Deployable, Auditable, Extensible

The pipeline is deterministic except for the Writer step. The same match data produces the same metrics and the same verification results every time. This auditability — the ability to trace every number in the report back to the source event it was computed from — is something no chatbot-based tool offers.

The custom JSON upload feature means any analyst with event data in Pitch Verdict format can immediately run the full pipeline on any match — not just the five built-in samples.

6.4 Challenges and Solutions

The hardest engineering challenge was the Verifier. Writing regex that reliably extracts numerical claims from natural language prose — and then correctly identifies which team each number refers to — took significant iteration. The model might say "Spain held 62% of the ball" or "possession was 62% in favour of Spain" or "the 62% possession figure reflects Spain's dominance" — all the same claim, written three different ways. The solution was layered: regex for the number, then a sliding context window of surrounding text, then team name frequency scoring to assign ownership. Tolerances (±2% for percentages, ±0.05 for xG) were calibrated through adversarial testing to minimise both false positives and missed errors.

The second challenge was keeping the LLM honest without constraining the prose. The first version of the Writer prompt let the model paraphrase metrics freely — "approximately two-thirds possession" instead of "65.3%". The Verifier could not match these. The fix was a stricter system prompt requiring exact numeric citation, which improved verification coverage from ~60% of claims to ~95% without meaningfully degrading narrative quality.

The third challenge was the multi-match architecture. Adding five different sample matches meant the synthetic data generator needed realistic per-match tactical profiles — not just random event distributions. Each match required hand-tuned possession rates, PPDA targets, and goal timings that reflected the actual published statistics from those games.

6.5 Future Improvements

The most impactful near-term addition would be formation detection — using positional clustering of event coordinates to infer the shape each team is playing at any given moment, rather than relying solely on PPDA and possession. This would allow the Tactical Classifier to identify shape changes mid-phase, not just between phases.

A second priority is seasonal context. Right now, every report treats the match in isolation. Integrating FBRef season averages would let the system say "Spain's PPDA of 5.8 is significantly below their season average of 8.2, suggesting they pressed unusually high for this fixture" — which is the kind of contextual insight analysts actually need.Real-time data integration is also on the roadmap. StatsBomb and Opta both offer live data feeds. With a streaming event endpoint, Pitch Verdict could generate and verify a phase report within minutes of a goal being scored, rather than post-match.

6.6 Ethical Considerations

All data used by Pitch Verdict is publicly available match performance data released by StatsBomb under their open data initiative. No private, personal, or sensitive information is used at any stage. Player names appear only in the context of on-ball events (goals, substitutions) and are sourced from public match records.

StatsBomb's user agreement requires source attribution. Every report generated by the system includes the line: *"Data sourced from StatsBomb Open Data. xG values use the StatsBomb xG model."* This is enforced structurally — the Writer's system prompt includes it as a mandatory closing line.

The Verifier handles a subtler ethical obligation: intellectual honesty. If a metric cannot be computed from the available data, the system outputs "data not available" rather than fabricating a plausible value. This is not just good engineering — it is a design commitment to not mislead the reader.

One limitation worth naming explicitly: PPDA and xG are models, not ground truth. They are widely used and respected, but they carry assumptions. A PPDA computed from event data misses off-ball pressing intensity. An xG model trained on one dataset may produce different values than one trained on another. Every report should be read as data-informed analysis, not objective fact — and Pitch Verdict's verification system checks internal consistency, not ground truth correctness.

Finally, the adversarial testing framework in the app demonstrates something important about responsible AI deployment: claims about system reliability should be testable, not assumed. The Adversarial Test tab exists precisely so users — including the professor grading this — can verify the verification system's performance themselves rather than taking it on faith.

7. Technical Stack

Component	Technology	Notes
Language	Python 3.10+	Industry standard for data science and ML pipelines
LLM	xAI Grok (grok-3-mini)	Free-tier API via OpenAI-compatible endpoint
Data	StatsBomb Open Data	Free, GitHub-hosted, used by Premier League clubs
Verification	Custom Python (deterministic)	Regex extraction + ground truth lookup, no LLM
UI	Streamlit	Rapid development, data-science native framework
Charts	Plotly	Interactive phase analysis visualisations
Deployment	Streamlit Community Cloud	Free hosting, live public URL
Cost per report	< $2.00	$0 data + ~$0.50–$2.00 LLM API calls

8. Limitations

- Off-ball blindness: StatsBomb event data captures on-ball actions only. Pressing runs that do not win the ball, decoy movements, and passive defensive shape are not captured. Every report acknowledges this when drawing off-ball conclusions.
- xG model dependency: xG values use StatsBomb's proprietary model. Different providers produce different values for the same shot. Every report discloses the model used.
- Verification tolerance: The ±2% tolerance on percentages prevents false positives from floating-point arithmetic but means very small deviations are accepted without flagging.

9. The Broader Idea

The generate-then-verify pattern Pitch Verdict demonstrates is not specific to football.

Medical report generation from clinical notes. Financial analysis from SEC filings. Legal research summaries for non-specialists. In every case the same question applies: when the model gives you a number, how do you know it is correct?

The standard answer in the AI industry is: it usually is, and we accept some error rate. Pitch Verdict proposes a different answer — build the verification into the pipeline. Make accuracy measurable, not assumed. Make every claim traceable back to the source it was computed from.

> *Pitch Verdict is not ultimately a football analytics tool. It is a demonstration that analytical AI systems can hold themselves accountable — if we design them to.*

10. Setup & Usage

Quick Start

```
git clone https://github.com/YOUR_USERNAME/pitch-verdict
cd pitch-verdict
pip install -r requirements.txt
cp .env.example .env
# Add XAI_API_KEY=gsk_your_key to .env
streamlit run app.py
```

Deployment

Live URL: **https://pitchverdict.streamlit.app** | The API key is stored as a Secret in Streamlit Cloud — never in the repository.